HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed advanced computer engineering challenges focusing on **RISC-V architecture** and **memory management**, creating detailed technical solutions and architectural diagrams.
- Authored rigorous, step-by-step technical solutions for AI training, including **assembly code snippets** and **hardware description language (HDL)** examples for embedded systems.
- Evaluated AI-generated code (**C/C++**) and logic gate designs for technical accuracy, efficiency, and adherence to industry standards in **digital logic design**.
- Identified logical fallacies in AI reasoning, such as **race conditions** and **memory leaks**, providing structured feedback to refine model accuracy.
- Developed Python scripts to automate data extraction and analysis for technical documentation, improving report generation efficiency by **~30%**.
- Containerized applications using **Docker** and deployed them to **AWS EC2** instances, managing CI/CD pipelines with **Jenkins** for consistent builds.
- Supported enterprise Java applications in production environments, managing SLAs and responding to P1/P2 incidents using **Splunk** and **Dynatrace**.

SKILLS

Core Engineering – Computer Architecture, Embedded Systems, Digital Logic Design, Operating Systems, Hardware-Software Co-design, RISC-V, ARM, FPGA Development, Memory Management, Hardware Security, Systems Software

Programming Languages – Python, Java, C/C++, Assembly, Verilog, VHDL, R

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Data Analysis & ML – pandas, NumPy, scikit-learn, Power BI, Tableau, DAX, Data Modelling, ETL, SQL

Tools & Technologies – Jira, Confluence, Linux, Shell Scripting, Postman, FastAPI

PROJECTS

Embedded Systems Challenge Generation Platform

- Developed a Python-based platform to generate complex engineering challenges for embedded systems, focusing on **memory management** and **real-time constraints**.
- Authored **assembly code** and **Verilog HDL** snippets as ground-truth solutions for AI training, covering topics like interrupt handling and peripheral interfacing.
- Integrated a basic scoring mechanism to evaluate AI-generated solutions against predefined correctness criteria and efficiency benchmarks.

Hardware Logic Design Evaluation Framework

- Built a framework using Python to analyze and evaluate AI-generated **Verilog/VHDL** code for digital logic designs, identifying potential issues like **race conditions**.
- Developed parsers to extract key design elements from HDL files, enabling automated checks for adherence to specific architectural patterns and timing constraints.
- Created sample testbenches and documentation to serve as reference examples for training AI models on hardware design principles.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)